

NEWSBOT INTELLIGENCE SYSTEM

Introduction

The NewsBot Intelligence System was developed to automate the classification and analysis of news articles.

Modern organizations consume large quantities of textual information daily, making automated processing an essential capability.

This project integrates multiple NLP techniques into a unified pipeline capable of extracting meaningful insights from unstructured news data.

Methodology

The project utilized a news article dataset obtained from Kaggle. Articles were preprocessed through tokenization, stopword removal, normalization, and lemmatization. TF-IDF vectorization transforms text into numerical features suitable for machine learning models.

Part-of-speech tagging and dependency parsing were used to analyze grammatical structures and writing patterns. Sentiment analysis measured emotional tone across categories. Named Entity Recognition identifies important entities such as people, organizations, locations, dates, and monetary values.

Three machine learning models were evaluated: Logistic Regression, Random Forest, and Multinomial Naive Bayes. Model performance was assessed using classification accuracy and prediction consistency.

Results

The Random Forest classifier achieved the strongest performance among tested models.

Technology articles contained the highest concentration of organization entities, while political articles demonstrated higher frequencies of person and geographic entities.

Sentiment analysis revealed noticeable differences between categories, with sports articles exhibiting more positive sentiment overall.

Business Value

NewsBot demonstrates how NLP can automate content classification, support media monitoring, and improve business intelligence operations.

Organizations can use such systems to identify trends, monitor public sentiment, and extract valuable information from large collections of articles.

Conclusion

This project successfully integrated all required NLP modules into a single system.

The results demonstrate the effectiveness of combining preprocessing, feature extraction, machine learning, sentiment analysis, syntax analysis, and entity recognition to transform raw text into actionable insights.